In summary, user data is a way to inject a script into the boot process of an EC2 instance—a simple but powerful concept. Note that by default the script is executed only during the first boot of the instance, so don’t rely on the script to start services.

Debugging a user data script

In case you need to debug a user data script, use the Session Manager to connect to the EC2 instance. Have a look at the /var/log/cloud-init-output.log log file shown next, which contains the outputs of the user data script at the end:

```
$ less /var/log/cloud-init-output.log
```

So far, we have deployed Etherpad on EC2 by injecting a bash script into the boot process with the help of user data. But how do we perform a rolling update to update Etherpad from version 1.8.17 to 1.8.18? See the next listing for details.

Listing 15.5 Updating the Auto Scaling group or the referenced launch template

```
AutoScalingGroup:                                    ①
  Type: 'AWS::AutoScaling::AutoScalingGroup'
  Properties:
    TargetGroupARNs:
    - !Ref LoadBalancerTargetGroup
    LaunchTemplate:                                  ②
      LaunchTemplateId: !Ref LaunchTemplate
      Version: !GetAtt 'LaunchTemplate.LatestVersionNumber'
    MinSize: '1'                                     ③
    MaxSize: '2'                                     ④
    HealthCheckGracePeriod: 300
    HealthCheckType: ELB
    VPCZoneIdentifier:
    - !Ref SubnetA
```